

Answer Key to Sec 3 6092 EYE 2023

Qn	Ans	Explanation
1	B	Volume of hydrogen gas collected is about 80cm ³ . Hence, a 100 cm ³ gas syringe is needed.
2	C	At point C, in the condenser, vapour cools to liquid and flows out as distillate.
3	B	In experiment 1, helium has a lower molecular mass than oxygen, hence helium will diffuse into the beaker faster than oxygen diffuse into the porous tube and liquid at point P rises. In experiment 2, helium has a higher molecular mass than hydrogen, hence hydrogen will diffuse into the porous tube faster than helium diffuse into the beaker and liquid at point Q is higher.
4	D	Products are a mixture of solid and aqueous solution. Hence, filtration is used to separate the products.
5	D	R _f value is calculated as the distance moved by spot (S-T) divided by distance moved by solvent (U-T)
6	B	As distillation proceeds, the concentration of ethanol decreases and eventually reached zero as ethanol is collected as a distillate (ethanol has a lower boiling point than water).
7	D	Both particles have more electrons (referencing the electronic configuration) than protons. S has 5 valence electrons while T has 6 valence electrons. Hence, both particles are ions of non-metals.
8	B	Relative atomic mass of Z = (0.5x90) + (0.1x91) + (0.2 x 92) + (0.2 x 94) = 91.3
9	A	Sodium ion has 10 electrons after giving away 1 electron to form a positive ion. Both sodium ion and neon have 10 electrons.
10	D	The number of electrons available to form bonds are dependent on the number of unpaired valence electrons. Oxygen has 2 unpaired valence electrons to form bonds, fluorine has 1 unpaired valence electron while carbon has 4 unpaired valence electrons.
11	A	V is located in group 15 while W having (n+2) protons, is in group 17. Both V and W are non-metals, hence form covalent compound. V has 5 valence electrons and W has 7 valence electrons; hence formula is VW ₃ .
12	B	W has 1 valence electron (H) while Y has 4 valence electrons (C), and Z has 6 valence electrons (Z).D
13	D	Substance O conducts electricity in both solid and liquid state, and only metals conduct electricity in both states.
14	D	% by mass of water = (5 x 18 x 100%) / 160 + (5 x 18)
15	D	Compound formed between Y and Z is Y ₂ Z. 1 mol of Y ₂ Z = (19 x 2) + 16 = 54.
16	A	Aluminum transfers three moles of electrons as it has three valence electrons.
17	A	XOH/H ₂ SO ₄ = (20 x 1) / (40 x 0.25) = 2 2 mol XO ₂ → 1 mol H ₂ SO ₄
18	C	1 mol CH ₄ → 2 mol O ₂ 20 cm ³ CH ₄ → 40 cm ³ O ₂ → 20 cm ³ CO ₂ → 40 cm ³ H ₂ O
19	C	No of moles of CO ₂ = 0.09 / 24 = 0.00375 mol 1 mol CO ₂ → 1 mol CaCO ₃ mass of CaCO ₃ = 0.00375 mol x 100 g = 0.375 g % purity = 0.375 g / 0.5 g = 75%
20	B	aqueous hydrogen chloride is a strong acid; hydrochloric acid.

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21	A	All acids have pH less than 7. Weak acids do not completely ionised in water. Unreactive metals do not produce hydrogen gas. Dibasic weak acids are weaker than monobasic strong acids.
22	C	Sulfur dissolves to form acid (pH2) while calcium dissolves to form alkali (pH 13). Aluminum oxide is white in colour.
23	D	Z will both turn indicators yellow as it has pH 6.
24	B	Copper has no reaction with hydrochloric acid. Silver nitrate and hydrochloric acid produces an insoluble salt, silver chloride.
25	D	At the start of experiment, pH is below 7 (acidic), indicating that solution Z is a weak acid. When solution Y is added, pH increases. When excess Y is added, pH is 14. Hence, this indicates that Y is a strong alkali.
26	D	As the volume of hydrochloric acid used increases in each titration, it is deduced that conical flask was washed with potassium hydroxide solution.
27	C	When dilute sulfuric acid is added to barium hydroxide, barium sulfate (insoluble ppt) is formed. Hence, the number of mobile ions decreases. When excess dilute sulfuric acid is added, the number of mobile ions increases as acids ionise in water.
28	C	Magnesium sulfate is a soluble salt and is prepared by reacting magnesium with hydrochloric acid.
29	B	When excess magnesium carbonate is added to dilute sulfuric acid, magnesium sulfate, water and carbon dioxide is produced. In step 2, unreacted magnesium carbonate is removed. In step 3, when heating the filtrate, water is removed to form crystals.
30	B	Oxidation state of Cl: Cl_2 : 0 NaCl: -1 NaClO ₃ : +5
31	C	No change to the oxidation states of elements before and after the reaction.
32	D	KI is a reducing agent, hence will change colour in the presence of an oxidising agent (Q).
33	B	When barium nitrate is added to calcium chloride, colourless solution is observed. When barium nitrate is added to sodium sulfate, a white precipitate (BaSO ₄) is produced.
34	A	Rubidium is an alkali metal. Hence, it does not have a high melting point like other metals.
35	D	Going down a group, the number of protons and electrons increase. Hence, the total number of electrons in the inner shells also increase.
36	D	Astatine is located at the bottom of group 17. Reactivity decrease down the group (astatine is less reactive than iodine) and boiling point increases down the group (astatine has a higher melting point than iodine). Hence, statements 1 and 2 are incorrect.
37	C	When the mass of lead decreases, this indicates that lead displaces the metallic ion from its solution. Hence, R is the most reactive as lead is unable to displace R from its solution (no change to mass). Lead is more reactive than both P and Q. Q is the least reactive as lead is able to displace Q from its solution (loss of 3.5 g).

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38	B	As the carbonate of Z did not decompose, this indicates that Z is more reactive than copper. Copper does not react with acid as it is an unreactive metal.
39	A	Both statements are correct but statement 1 does not explain statement 2.
40	C	Both silver and zinc are below carbon in the reactivity series, hence, carbon can be used to reduce the oxides of two metals. Magnesium is positioned above carbon, hence, cannot be reduced by carbon.

Answer Key to Sec 3 6092 EYE 2023

Qn number	Topics	need to revise (✓)
1	Experimental Chemistry	
2	Kinetic Particle Theory	
3	Kinetic Particle Theory (Diffusion)	
4	Methods of Purification	
5	Methods of Purification (Paper Chromatography)	
6	Methods of Purification (Fractional Distillation)	
7	Atomic Structure	
8	Atomic Structure (Relative Abundance)	
9	Atomic Structure	
10	Chemical Bonding	
11	Chemical Bonding (Covalent)	
12	Chemical Bonding (Covalent)	
13	Chemical Bonding (Structures)	
14	Mole Concept	
15	Mole Concept	
16	Mole Concept	
17	Mole Concept (Concentration)	
18	Mole Concept (Limiting Reagent)	
19	Mole Concept (Percentage Purity)	
20	Acids and Bases	
21	Acids and Bases	
22	Acids and Bases (Oxides)	
23	Acids and Bases (indicators)	
24	Acids and Bases (Acid Reactions)	
25	Acids and Bases (pH curves)	
26	Salts (Titration)	
27	Salts (Precipitation)	
28	Salts	
29	Salts	
30	Redox Reactions	
31	Redox Reactions	
32	Redox Reactions	
33	Qualitative Analysis	
34	Periodic Table	
35	Periodic Table	
36	Periodic Table	
37	Metals (Reactivity of Metals)	
38	Metals (Metal Decomposition)	
39	Metals (Alloys)	
40	Metals (Extraction of Metals)	